Neuroimaging: Methods, Findings, and Limits

Chapter 13

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Central question. What can brain imaging tell us about trauma, and what can it not tell us?

Learning objectives

- describe the main neuroimaging methods and what each measures;
- explain the difference between structure, function, connectivity and chemistry;
- evaluate replicated findings in trauma neuroimaging;
- recognise common misinterpretations of imaging results;
- explain the limits of imaging as a diagnostic tool.

Introduction

Neuroimaging has transformed the study of trauma by allowing researchers to observe the living brain. It has also generated some of the most misleading claims in the field. This chapter explains what imaging methods measure, summarises replicated findings, and sets out the limits of inference.

The central point is that imaging measures biological signals that are related to brain activity indirectly. Functional magnetic resonance imaging, for example, measures changes in blood oxygenation, not neuronal firing. Interpreting those signals requires assumptions, models and statistics, and the results depend on the task, the sample, the analysis pipeline and the comparison group.

> **A note on brain images.** *A brain image in a news article is usually a statistical map overlaid on an anatomical template. Coloured areas indicate where a statistical test exceeded a threshold. They do not show "activity" in the way a photograph shows a face, and a difference between groups does not establish cause, diagnosis or individual prediction.*

13.1 Structural imaging

Structural imaging includes computed tomography and magnetic resonance imaging. CT is fast and useful in acute settings for detecting haemorrhage, fracture and mass lesions. MRI provides higher resolution and better soft-tissue contrast, and is used for volumetry, cortical thickness and white-matter imaging.

Structural studies in PTSD have reported reduced hippocampal volume, reduced anterior cingulate and orbitofrontal cortex volume, and altered

amygdala volume. Effect sizes are modest, findings are heterogeneous, and direction of causality is unclear. Childhood adversity, depression, alcohol use and medication may all contribute.

Importantly, group differences do not translate into individual diagnosis. The distributions overlap substantially, and no structural measure can identify a person with PTSD.

13.2 Functional MRI

Functional MRI measures blood-oxygen-level-dependent (BOLD) signal, which reflects changes in deoxygenated haemoglobin and is coupled to neural activity through neurovascular mechanisms. It has good spatial resolution and moderate temporal resolution.

Task-based fMRI compares activity during different conditions. In trauma research, common paradigms include fear conditioning, extinction recall, emotional face processing, symptom provocation and working-memory tasks. Results depend heavily on the paradigm, and a finding in one task may not generalise.

Resting-state fMRI measures spontaneous fluctuations in BOLD signal and is used to estimate functional connectivity between regions. It is attractive because it does not require a task, but connectivity estimates are sensitive to preprocessing, motion and physiological noise.

13.3 Positron emission tomography

PET uses radiotracers to measure metabolic activity or receptor availability. Fluorodeoxyglucose PET measures glucose metabolism; ligand-based PET can measure receptors, transporters or immune markers.

PET has lower spatial and temporal resolution than MRI and involves radiation exposure. It is used mainly in research and in specific clinical questions, such as distinguishing neurodegenerative conditions. In trauma research, PET has been used to study receptor systems and, more recently, microglial activation.

13.4 Connectivity and networks

Connectivity analyses examine how regions co-vary or interact. Functional connectivity is typically estimated from correlations in time series; effective connectivity attempts to model directed influence.

Graph-theoretic approaches describe network properties such as modularity, efficiency and hub structure.

The triple-network model describes interactions among the default mode network, the salience network and the central executive network (Menon, 2011). In PTSD, studies have reported altered coupling among these networks, but findings vary with analysis method and sample.

Connectivity findings should be interpreted cautiously. Correlation does not establish direction, and network boundaries are defined by analysis choices rather than by anatomical borders.

13.5 Replicated findings in trauma

Several findings have been reported repeatedly, though with heterogeneity. The amygdala tends to show heightened reactivity to threat-related stimuli (Etkin & Wager, 2007). The ventromedial prefrontal cortex tends to show reduced activation and reduced connectivity with the amygdala. The hippocampus shows altered structure and function, particularly in relation to contextual processing and memory.

The dissociative subtype shows a different pattern, with heightened prefrontal control over limbic regions rather than reduced control (Lanius et al., 2010). This is an important corrective to the assumption that PTSD always involves "under-regulation".

These findings are group-level and probabilistic. They do not describe any individual patient, and they should not be used to justify claims about a person's brain based on a single scan.

Recent reviews of the imaging literature in adults and youth reach broadly similar conclusions about amygdala and prefrontal involvement, while noting persistent methodological limitations (Hinojosa et al., 2024).

13.6 Fear circuitry

The fear circuitry model emphasises the amygdala, hippocampus, insula, anterior cingulate and prefrontal cortex. The amygdala detects salience and supports threat learning; the hippocampus provides context; the insula represents bodily state; the anterior cingulate integrates conflict, pain and affect; the prefrontal cortex regulates responses.

This model is useful for organising findings but should not be reified. Fear is not located in a single region, and the same regions participate in many other functions.

13.7 Childhood maltreatment

Neuroimaging studies of childhood maltreatment report structural and functional differences, including reduced hippocampal and corpus callosum volume, altered amygdala reactivity and altered prefrontal connectivity. Teicher and Samson (2016) reviewed this literature and estimated that maltreatment accounts for a substantial proportion of the population attributable risk for childhood-onset psychiatric disorders.

The interpretation is complicated by confounding: maltreatment often co-occurs with poverty, neglect, family instability and other adversities. Studies that attempt to isolate maltreatment effects find smaller and more specific differences than studies that compare broadly disadvantaged groups.

13.8 Predictive modelling

Machine-learning approaches have been used to classify individuals as having PTSD versus not, or to predict treatment response. Reported accuracies are often inflated by small samples, overfitting and lack of independent validation.

For a biomarker to be clinically useful, it must add value beyond existing assessment, be validated in diverse populations, and have acceptable cost and burden. No neuroimaging biomarker currently meets these criteria for PTSD.

13.9 Publication bias and replicability

Neuroimaging has well-documented problems with replicability. Small samples produce unstable estimates; flexible analyses allow many possible results; selective reporting of significant findings is common; and effect sizes in published studies tend to be larger than in replication attempts.

Larger consortia, preregistration, data sharing and multiverse analyses are improving the situation. Readers should check sample size, replication status and whether the analysis was preregistered before accepting a finding.

13.10 Ethical communication

Imaging findings are persuasive to the public and to courts. They can be used to argue that a person is damaged, that they are not responsible, or that their symptoms are real. These uses may sometimes serve justice, but they can also reinforce stigma and biological determinism.

Communicating imaging research responsibly means avoiding claims that exceed the data, explaining group-level findings, and resisting the temptation to localise complex experiences to a single region. It also means involving people with lived experience in how findings are represented.

13.11 Designing an imaging study

An imaging study begins with a question that imaging can answer. "What does trauma do to the brain" is not a research question; "do adults with PTSD show greater amygdala response to masked threat faces than trauma-exposed controls without PTSD" is.

Power is the first constraint. Effect sizes in psychiatric imaging are small, and underpowered studies produce unstable and inflated estimates. Sample sizes in the dozens are common and usually inadequate for anything beyond a large effect. Power calculations should be reported, and they should be based on realistic effect sizes rather than optimistic ones.

Comparison groups determine interpretation. Healthy controls differ from trauma-exposed people in many ways beyond diagnosis, so trauma-exposed controls without PTSD are often more informative. Comorbidity, medication, substance use, sleep and motion must be measured and, where possible, modelled.

Task design matters. A task that does not reliably engage the process of interest will produce null or ambiguous results. Pilot testing, manipulation checks and reporting of behavioural data are essential; a group difference in brain activation is uninterpretable if the groups also differ in performance or effort.

Analysis choices multiply degrees of freedom. Preprocessing pipelines, smoothing, motion correction, region definitions, contrast specification and multiple-comparison correction all affect results. Preregistration, analysis code and reporting of the full pipeline reduce the risk that a finding reflects the analyst rather than the brain.

13.12 Multimodal and longitudinal designs

No single imaging method captures the brain. Structure, function, connectivity, chemistry and blood flow provide different views, and combining them can constrain interpretation. A structural difference without a functional correlate, or a connectivity difference without a behavioural one, is a hypothesis rather than a mechanism.

Multimodal studies are expensive and analytically demanding. Combining datasets requires attention to registration, scaling and missing data, and the risk of spurious findings increases with each additional analysis.

Longitudinal designs are more informative than cross-sectional ones because they can distinguish pre-existing differences from consequences of trauma. They are difficult: attrition is often related to symptoms, repeated scanning is burdensome, and the interval between assessments may miss the relevant change.

Consortia and data-sharing initiatives have increased sample sizes and improved reproducibility. They have also revealed how much variation exists across sites, scanners and populations, and how poorly some findings generalise. The ENIGMA-style approach—harmonised analysis of many cohorts—is now standard for genetic and structural questions, though it cannot fix weak measurement.

For trauma research, the most valuable longitudinal questions are about prediction and mechanism: which responses shortly after an event predict later symptoms, and which changes accompany recovery. Answering them requires repeated measurement, low attrition and clear prespecification.

Synthesis

Table 13.1 summarises imaging methods.

Table 13.1. Neuroimaging methods and their limits

Method	Measures	Strengths	Limits
CT	X-ray attenuation	Fast, accessible	Radiation, low soft-tissue contrast
Structural MRI	Anatomy	High resolution	Correlation not causation
Task fMRI	BOLD during task	Spatial localisation	Indirect, task-dependent
Resting-state fMRI	Spontaneous BOLD	No task required	Sensitive to motion and preprocessing
PET	Metabolism, receptors	Molecular specificity	Radiation, cost, low resolution
EEG/MEG	Electrical/magnetic fields	Millisecond timing	Poor spatial resolution

Composite case study

A patient brings a private fMRI report claiming "amygdala hyperactivity consistent with trauma". The clinician explains that group-level findings cannot be applied to an individual, that the scan does not establish diagnosis, and that the report should not be treated as evidence in a legal dispute without expert review. The clinical assessment proceeds on the basis of history and symptoms.

Review and discussion questions

What does BOLD signal actually measure?

Why do group differences not support individual diagnosis?

How does the dissociative subtype complicate the fear-circuitry model?

What are the main sources of irreplicability in imaging research?

How should imaging evidence be presented in legal settings?

What would a clinically useful biomarker need to demonstrate?

Research assignments

- Critically appraise one task-fMRI study of PTSD, noting sample size and correction methods.

- Design a preregistered replication of an amygdala reactivity finding.
- Write guidance for journalists on reporting brain imaging studies.

Chapter summary

Neuroimaging provides indirect measures of brain structure, function, connectivity and chemistry and has identified replicable group-level differences in trauma. These findings have advanced mechanistic understanding but do not support individual diagnosis, and the field faces serious replicability challenges. Responsible communication requires accuracy about what the methods measure, humility about inference, and resistance to biological determinism.

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